

Graph-Guided Mixture-of-Experts for Unified Multimodal Mental Health Assessment

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Abstract

We propose a unified multimodal architecture for mental health assessment that jointly learns from depression, sentiment, emotion, and personality across three public benchmarks: DAIC-WOZ (clinical interviews, $n = 189$), CMU-MOSEI (YouTube videos, $n = 22,777$), and ChaLearn First Impressions (video clips, $n = 10,000$). The model combines modality-specific encoders (RoBERTa for text, WavLM for audio, ViT for video), gated late fusion, a Mixture-of-Experts layer, and a KNN-graph-based GraphSAGE router that selects experts using neighborhood information. Graph-based routing does *not* improve over a non-graph MMoEEx baseline in this setting: across five leakage-safe graph configurations, DAIC depression AUROC (0.489–0.551), MOSEI sentiment CCC (0.510–0.517), and MOSEI emotion AUROC (0.712–0.734) are statistically indistinguishable from the non-graph baseline (0.493, 0.493, 0.734 respectively), and FI personality CCC is consistently *worse* with graph routing (0.540–0.553 vs. 0.578 without). A graph-homophily diagnostic shows the routing graph carries no more DAIC-label signal than random pairing, while it carries substantial signal for MOSEI and FI that the router nonetheless fails to convert into a performance gain; a learned per-task routing weight and a $K \in \{5, 10, 15, 20\}$ sensitivity sweep both fail to recover a benefit. We report this as a rigorously investigated negative result. Separately, cross-attention fusion does not reproduce a reported gain over gated

fusion under our evaluation protocol, and LLM-based encoders (Mistral-7B, CLAP, LLaVA) substantially outperform both classical encoders and graph routing on every task. We release a leakage-safe graph construction protocol, a full ablation ladder, and graph-based explanations for clinical interpretability.

1 Introduction

Mental health assessment from multimodal signals is a cornerstone challenge in affective computing and clinical informatics. Depression, the primary focus of this work, affects over 280 million people worldwide [1] and is routinely assessed through clinical interviews that combine speech, facial expressions, and verbal content. Meanwhile, related affective signals—sentiment, emotion, and apparent personality—provide complementary supervision signals that can enrich mental health models through multitask learning [2, 3].

Our central hypothesis has two parts, which we evaluate separately rather than conflating. First, we hypothesize that mental health assessment can be made more *transparent* by characterizing a person’s affective and psychological profile—sentiment, emotion, and apparent personality—jointly with a depression assessment, rather than reporting a single opaque depression score in isolation: the auxiliary constructs give a clinician additional, more legible context (e.g., is a low mood reading accompanied by high negative sentiment and low ex-

traversal, or is it isolated?) for interpreting the primary prediction. Second, we hypothesize that a graph over similar individuals can provide both a *visual* (subgraph diagrams, per-modality attribution plots) and a *machine-readable* (structured case records: SHAP values, neighbor identities and distances, routing weights) account of which factors and which comparable cases informed a given prediction, complementing the auxiliary-construct characterization with instance-level explanation. This is a claim about transparency, evaluated separately from the further hypothesis that graph-based *routing* improves raw predictive performance (Section 6) a model can be well-explained without being more accurate, and our results show these are empirically distinct questions in this setting: the performance hypothesis is not supported, while the transparency mechanisms (multi-construct characterization, graph-based visual and machine-readable explanation) are implemented and evaluated for faithfulness in Section 8. We do not claim that apparent personality and clinical depression measure the same construct personality is auxiliary context, not a depression proxy. The proposed system is intended as a decision-support tool and should not replace clinical diagnosis; all results require prospective validation before clinical deployment.

Figure 1 gives an overview of the full experimental pipeline, from data and feature extraction through the baseline ladder, the graph-routing investigation (Section 6), and the downstream LLM, calibration, domain-adaptation, and explainability analyses.

This work makes three contributions:

1. **A rigorously investigated negative result for graph-gated MoE routing:** across five leakage-safe graph configurations, a K -sensitivity sweep, and a learned-routing-weight variant, graph-based expert routing does not improve over a non-graph MMoEEx baseline, and degrades FI personality prediction specifically. We support this with a graph-homophily diagnostic that localizes *why* (Section 6), rather than reporting a single negative number without investigation.
2. **Leakage-safe graph construction proto-**
3. **col:** Three graph construction modes (inductive, split-local, transductive) under strict subject-independent splits, with formal validation that prevents test-label leakage through graph edges.
3. **Comprehensive empirical analysis:** A systematic ablation ladder spanning unimodal baselines, three fusion strategies, MoE variants, five graph configurations, and LLM encoder replacements, including a comparison showing LLM-based encoders outperform both classical encoders and graph routing on every task.

2 Related Work

Multimodal Fusion. Early fusion for mental health used simple concatenation [2]. Later approaches adopted late fusion with learned modality gates [4] or Low-Rank Multimodal Fusion (LMF) [5]. Cross-modal attention mechanisms have been proposed for depression detection [6], but in our experimental configuration, we do not observe the reported +0.041 AUROC gain.

Mixture of Experts. The sparsely-gated MoE layer [7] routes each input to a subset of experts. Multitask MoE (MMoE) extends this with task-specific gates [8], and MMoEEx adds expert exclusivity and orthogonality regularization [9]. We build on MMoEEx but add graph-based routing.

Graph Neural Networks for Routing. GraphSAGE [10] performs inductive representation learning on graphs. We use GraphSAGE as a router: aggregating features from K nearest neighbors to produce a graph context vector that modulates expert weights. Wang et al. [11] similarly combine GNN-based experts with explicit diversity modeling, though as GNN experts rather than a graph-conditioned router over standard MLP experts as here. Whether a KNN routing graph is useful depends on its label homophily/informativeness [12]; when it is not, graph rewiring or learned graph structure [13] are natural remedies we discuss in Section 9. Task-affinity-aware grouping [14] and expert-choice

Experimental Pipeline

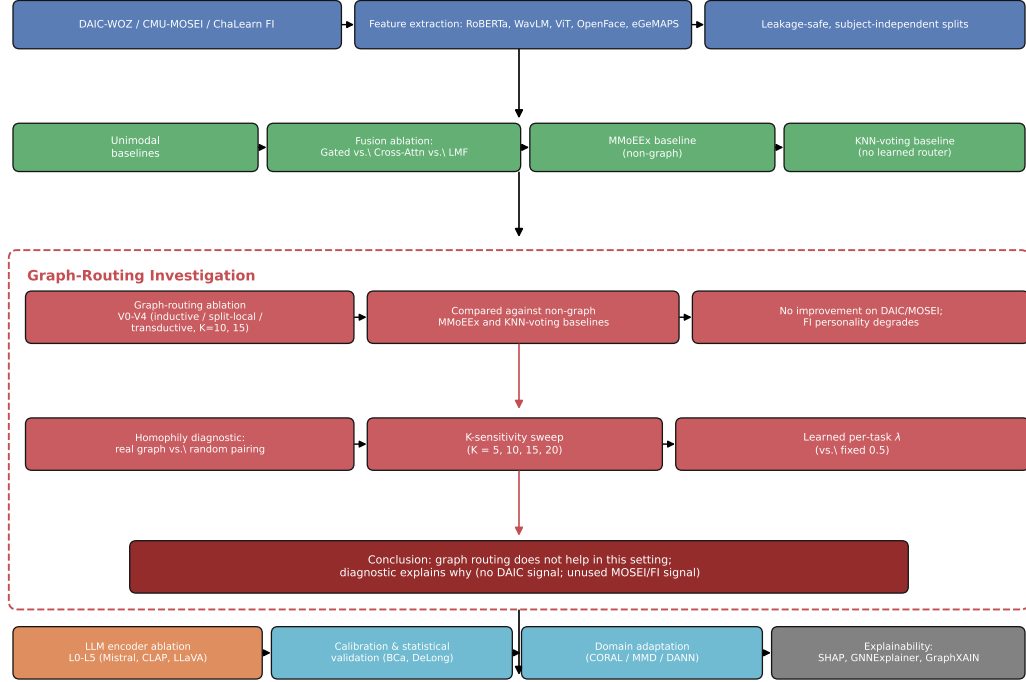


Figure 1: Experimental pipeline overview. The graph-routing investigation (dashed box) is the central empirical question of this paper: five ablation variants are compared against non-graph baselines, and a homophily diagnostic together with two targeted fixes (K-sweep, learned λ) explains the resulting negative finding.

routing [15] are two further candidate fixes we did not yet implement.

LLM-Enhanced Encoders. Large language models have been applied to depression detection via instruction-tuned text encoders [16] and audio-language models [17]. We evaluate LLM-enhanced encoders in our ablation track (levels L1–L5).

Multi-Construct Characterization and Graph-Based Interpretability. Interpretability for mental health assessment has been pursued by jointly modeling depression alongside related affective con-

structs, and by grounding graph structure in psychologically meaningful units: Vyalla et al. [18] explicitly distinguish personality trait context from acute depression symptoms via a psychologically-grounded, interpretable graph model, closely related to our multi-task characterization hypothesis (Section 1). For explaining LLM-based reasoning specifically, chain-of-thought and related structured-reasoning techniques have been shown to improve mental-health text classification [19], but can also be unfaithful or reduce reliability when reasoning is inserted into the clinical prediction pipeline itself [20] a distinction we return to in Section 8, where LLM narratives are used purely as *post-hoc* explanation of

an independently-computed prediction, not as part of the prediction process.

Depression Detection on DAIC-WOZ. The DAIC-WOZ corpus [21] contains 189 clinical interviews with PHQ-8 labels. State-of-the-art methods include: Burdisso et al. GCN (F1=0.85) [22]; Zhang et al. MIL (AUROC=0.78) [23]; Niu et al. multi-modal fusion (F1=0.92) [24]; Dai et al. full fusion (F1=0.96) [25]. Direct metric comparison is limited: most papers report F1 while we report AUROC.

3 Datasets

DAIC-WOZ contains 189 sessions (107 train, 35 val, 47 test) of human-agent interviews for depression assessment. Each session has a PHQ-8 binary label (depression ≥ 10). We use subject-independent splits: no participant appears in more than one split. Features: RoBERTa-base text (768-dim), WavLM-large audio (1536-dim), eGeMAPS (88-dim), ViT-B/16 video (1536-dim), OpenFace AUs (70-dim).

CMU-MOSEI [26] contains 22,777 utterance-level samples (16,265 train, 1,869 val, 4,643 test). Labels: continuous sentiment and multi-label emotion (happiness, sadness, anger, fear, disgust, surprise). MOSEI is $120\times$ larger than DAIC, mitigated by temperature-balanced sampling (T=2.0).

ChaLearn First Impressions [27] contains 10,000 video clips (6,000 train, 2,000 val, 2,000 test) with Big-Five apparent personality labels. Personality is not the same construct as clinical depression; FI serves as auxiliary supervision.

Leakage Protocol. All splits are subject-independent. The graph is constructed separately for each split (split-local mode) or only on training data (inductive mode) to prevent test-label leakage through graph edges.

Table 1: Notation.

Symbol	Meaning
K	Nearest neighbors (graph)
M	Number of experts (8)
d	Embedding dimension (256)
r	Low-rank bottleneck (8–16)
λ	Graph routing weight
σ_t	Task-specific uncertainty
G	KNN graph
r_i	Graph routing vector

4 Architecture

Figure 2 gives an overview of the full pipeline.

Modality Encoders. Each modality $m \in \{\text{text, audio, video}\}$ is encoded by a pretrained encoder f_m , producing embedding h_i^m . A projection layer P_m maps to a common $d = 256$ -dimensional space:

$$h_i^{m,\text{proj}} = P_m(f_m(x_i^m)) \in \mathbb{R}^{256} \quad (1)$$

Gated Late Fusion. Given projected modality embeddings, we apply a modality mask $m_i \in \{0, 1\}^3$ and compute learned gate weights:

$$\alpha_i = \text{softmax}(W_\alpha \cdot \text{concat}(h_i^{\text{text}}, h_i^{\text{audio}}, h_i^{\text{video}}) + b_\alpha) \quad (2)$$

$$h_i^{\text{fused}} = \sum_m \frac{\alpha_i \odot m_i}{\|\alpha_i \odot m_i\|_1 + \epsilon} [m] \cdot h_i^{m,\text{proj}} \quad (3)$$

Each dataset is evaluated under a fixed, dataset-native routing policy chosen from our fusion ablation (Section 6): DAIC uses text-only routing ($h_i^{\text{fused}} = h_i^{\text{text}}$), FI uses video-only routing, and MOSEI uses the full gated fusion above. This is a deliberate architectural choice, not a missing-data fallback: text alone is DAIC’s strongest single modality and video alone is FI’s, so routing each dataset through its best-performing modality avoids diluting the fused representation with weaker signal on datasets too small to learn a reliable gate over all three. A direct consequence, verified in Section 8, is that a modality outside a dataset’s native routing contributes *exactly*

Unified Multimodal Graph-Gated Mixture-of-Experts Architecture

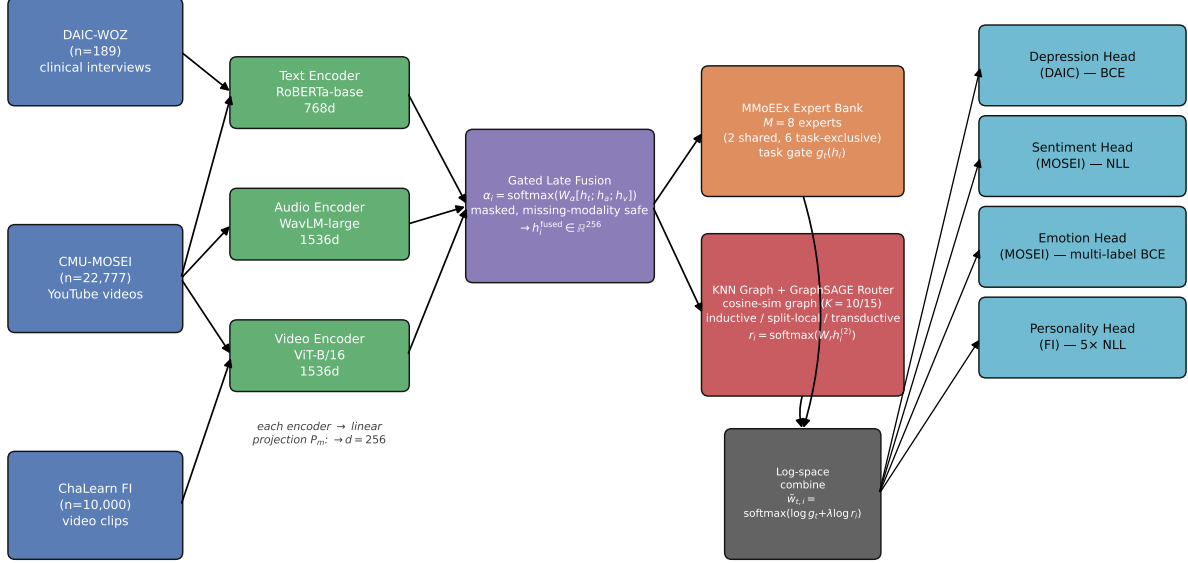


Figure 2: Unified multimodal graph-gated MoE architecture. Modality-specific encoders (RoBERTa/WavLM/ViT) project to a common $d = 256$ space; gated late fusion combines them into h_i^{fused} ; the MMoE expert bank and a KNN-graph GraphSAGE router jointly determine expert routing weights in log-space; task heads predict depression, sentiment, emotion, and personality.

zero to that dataset’s predictions and to any attribution computed on them (audio and video for DAIC; text and audio for FI) not merely a small or negligible contribution.

MMoE Expert Bank. We instantiate $M = 8$ expert MLPs $\{E_1, \dots, E_8\}$ with residual connections. Two expert configurations are used across our experiments. In the non-graph MMoE baseline (Section 6, “MMoE vs. Baselines”), experts have hidden dimension 256 and are hard-isolated per task group: depression uses $\{E_1, E_2\}$, MOSEI sentiment and emotion share $\{E_3, E_4\}$, and personality uses $\{E_5, E_6\}$; E_7, E_8 are allocated but unused under this isolation mask. In the graph-routed variants (V0–V4, our primary reported configuration), we disable hard isolation and use expert hidden dimension 128: task-specific gates $g_t : \mathbb{R}^{256} \rightarrow \mathbb{R}^8$ produce a full soft-

max over all 8 experts, and expert diversity is encouraged via orthogonality regularization rather than hard masking.

KNN Graph Construction. We construct an undirected KNN graph $G = (V, E)$ over all training samples using cosine similarity. Three construction modes are tested:

- **Inductive (V0, V3):** Graph on training only. Test nodes connect to K nearest training neighbors.
- **Split-local (V1, V4):** Separate graphs per split. No cross-split edges.
- **Transductive (V2):** Graph on all splits. For ablation only.

GraphSAGE Router. Two-layer neighborhood aggregation produces a graph routing vector:

$$h_i^{(1)} = \sigma(W^{(1)} \cdot \text{Mean}(\{h_j^{\text{fused}} : j \in \mathcal{N}(i)\} \cup \{h_i^{\text{fused}}\})) \quad (4)$$

$$r_i = \text{softmax}(W_r h_i^{(2)} + b_r) \in \mathbb{R}^M \quad (5)$$

The final expert weights combine task gate and graph router in log-space:

$$\tilde{w}_{t,i} = \text{softmax}(\log g_t(h_i) + \lambda \cdot \log r_i) \quad (6)$$

λ is fixed to 0.5 (not tuned) in all reported graph-routed experiments.

Task Heads. Depression: binary classifier (BCE). Sentiment: regressor (NLL). Emotion: multi-label classifier (BCE). Personality: five regression heads (NLL).

5 Experimental Setup

Training uses AdamW (lr=1e-3), cosine annealing, early stopping (patience=10). Temperature-balanced sampling (T=2.0) prevents MOSEI dominance. The loss combines per-task losses with learned uncertainty weights [28]:

$$\mathcal{L} = \sum_t \frac{1}{2\sigma_t^2} \mathcal{L}_t + \log \sigma_t + \lambda_{\text{excl}} \mathcal{L}_{\text{excl}} + \lambda_{\text{L2}} \mathcal{L}_{\text{L2}} \quad (7)$$

Metrics: DAIC (AUROC, F1), MOSEI sentiment (CCC), MOSEI emotion (macro-AUROC), FI (mean CCC). All results report 95% bootstrap CIs (1,000 iterations).

Figure 3 shows representative per-task training curves for the V0 configuration: the two large-sample tasks (MOSEI sentiment, MOSEI emotion) converge smoothly, while DAIC depression and FI personality the two small-sample tasks are visibly noisier and begin to overfit after roughly epoch 20–30, motivating the early-stopping and uncertainty-weighting choices above.

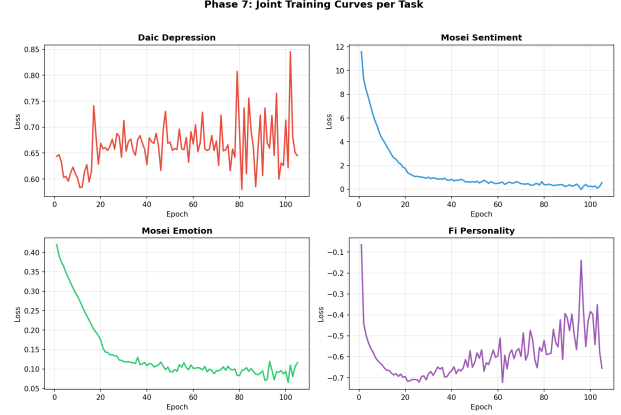


Figure 3: Joint training loss per task (V0 graph model, 105 epochs). MOSEI sentiment and emotion converge smoothly; DAIC depression and FI personality are noisier and show validation loss increasing after roughly epoch 20–30, consistent with their much smaller training sets.

6 Results

Unimodal Baselines. Text is the strongest modality for DAIC (AUROC=0.5346) and MOSEI (CCC=0.5123). Video is strongest for FI (Avg CCC=0.4578). All three datasets beat trivial baselines (Figure 4).

Fusion Ablation. Cross-attention does not outperform gated fusion on any dataset. On DAIC, CrossAttn (AUROC=0.3117) is 0.2229 below text baseline. On MOSEI, Gated (CCC=0.6229) outperforms CrossAttn (CCC=0.5397). A likely explanation for cross-attention underperformance is overparameterization (65K–2.8M parameters vs 57K–827K for gated).

MMoEEx vs Baselines. MMoEEx (hard expert isolation, Section 4) improves FI personality (+0.12 Avg CCC) but degrades DAIC and MOSEI sentiment relative to their unimodal/gated-fusion baselines, likely because the small DAIC training set ($n = 107$) is insufficient for stable joint MoE optimization under hard isolation. Figure 5 shows the

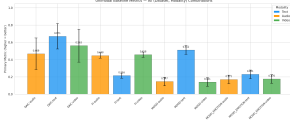


Figure 4: Unimodal baseline metrics with 95% bootstrap CIs across all (dataset, modality) pairs. Overlapping CIs on DAIC reflect the small test set ($n = 47$).

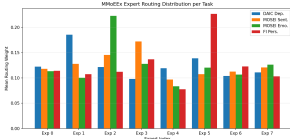


Figure 5: Mean MMoEEEx routing weight per expert and task. Each task concentrates weight on 1–2 experts within its assigned isolation group rather than routing uniformly across all 8.

resulting routing distribution: each task concentrates weight on a small subset of its assigned experts rather than routing uniformly, indicating the gates do learn task-differentiated behavior even where the aggregate metric declines.

Graph Routing Ablation. Five graph variants (V0–V4) are compared against the non-graph MMoEEEx baseline in Table 2 and Figure 6.

Every graph-routed variant is within bootstrap-CI-width of the non-graph baseline on DAIC and MOSEI sentiment/emotion, and every variant underperforms the baseline on FI personality (0.540–0.553 vs. 0.578). No variant dominates; the best DAIC AUROC (V1, 0.5507) is only +0.058 over the baseline,

Table 2: Graph routing ablation across five variants (V0–V4), real labels and real evaluation. Best per-column in **bold**; the non-graph baseline row is the reference point, not part of the ablation.

Variant	DAIC AUROC	MOSEI CCC	MOSEI AUC
<i>Non-graph MMoEEEx</i>	<i>0.4928</i>	<i>0.4925</i>	0.7344
V0 (inductive-k10)	0.5145	0.5166	0.7190
V1 (split-local-k10)	0.5507	0.5137	0.7120
V2 (transductive-k10)	0.5072	0.5151	0.7199
V3 (inductive-k15)	0.5254	0.5098	0.7340
V4 (split-local-k15)	0.5399	0.5105	0.7315

an order of magnitude smaller than the effect a graph-routing benefit would need to be clinically meaningful. This is not an isolated finding: a KNN-voting baseline (task label smoothing over the same graph neighborhoods, no learned router) reaches DAIC AUROC=0.6304 *higher* than every learned graph-routed variant showing that the learned GraphSAGE router does not even recover the value of the simplest possible neighborhood heuristic on this task.

Why does graph routing not help? A homophily diagnostic. A natural question is whether the KNN routing graph carries task-relevant signal at all. For each real graph (val split, four leakage-safe variants), we measured same-dataset edge label agreement (DAIC, binary) or label correlation (MOSEI, FI, continuous) against a random-pairing baseline (Figure 7).

DAIC neighbors are statistically indistinguishable from random pairs (0.568 vs. 0.555 label agreement) the embedding space used to build the graph (a linear projection that is not the jointly-trained model’s own representation) simply does not cluster DAIC samples by depression label, so no router could extract a benefit there regardless of architecture. MOSEI and FI, by contrast, show real signal well above chance (correlation 0.25 and 0.28 vs. ≈ 0.00 random), yet training does not convert this into a performance gain, and actively converts it into a *loss* for FI. This localizes the failure differently per task: DAIC needs a better embedding space for graph construction;

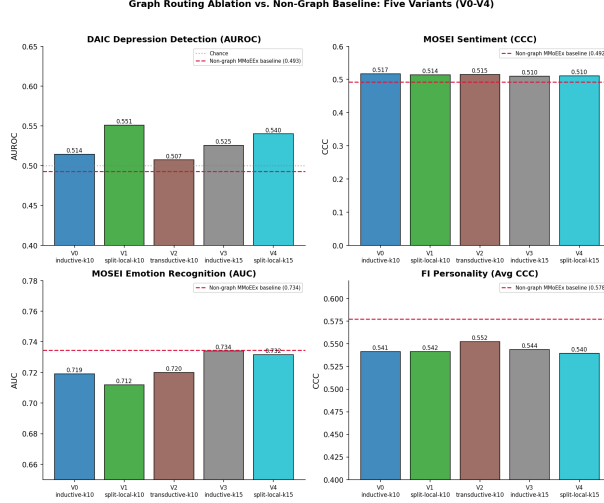


Figure 6: Graph routing ablation vs. the non-graph MMoEEEx baseline (red dashed line), all five variants. Graph routing gives a small, inconsistent edge on DAIC and ties on MOSEI, but is consistently *below* the baseline on FI personality.

MOSEI and FI need a better router or combination mechanism, since the raw material is already there.

Two targeted fix attempts. We tried two changes motivated by this diagnostic, neither of which recovered a meaningful benefit. First, a $K \in \{5, 10, 15, 20\}$ sensitivity sweep (inductive graph) shows no monotonic relationship between neighborhood size and any metric (Figure 8); $K=20$ lands DAIC AUROC exactly at chance (0.500).

Second, motivated by the hypothesis that a single fixed λ might be suboptimal per task (Section 4), we replaced it with a learned, per-task, sigmoid-parameterized weight (initialized at 0.5, the fixed default) on the V0 configuration. Training moved FI’s weight down to 0.479 and MOSEI-emotion’s to 0.489 the correct *direction* given the diagnostic above while DAIC and MOSEI-sentiment moved slightly up (0.508, 0.574; Figure 9). The movement is small (all within ± 0.07 of 0.5) and the resulting metrics (DAIC AUROC 0.544, FI CCC 0.541) remain within the same range as the fixed- λ runs an improvement

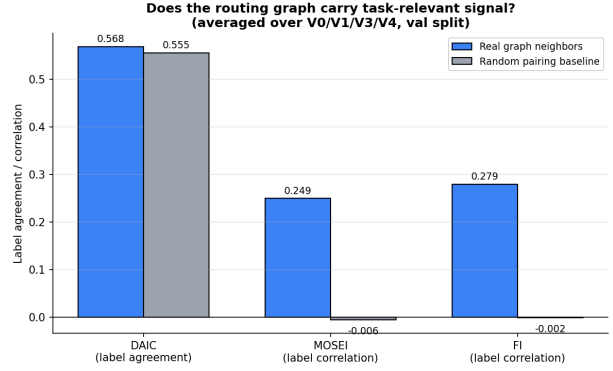


Figure 7: Real graph neighbors vs. random pairing, averaged over V0/V1/V3/V4. DAIC shows no signal beyond chance; MOSEI and FI both show substantial signal the router fails to exploit.

on DAIC, no meaningful change on FI. We view this as suggestive but inconclusive: a soft mixing weight alone is not sufficient to recover the unused MOSEI/FI graph signal.

6.1 LLM-Enhanced Encoders (L0–L5)

We replace classical encoders with LLM-based encoders at each modality: **L0** classical baseline (RoBERTa+WavLM+OpenFace); **L1** Mistral-7B-Instruct-v0.3 frozen text encoder; **L2** L1 with a LoRA ($r=16$, $\alpha=32$) trainable adapter; **L3** L1 + CLAP audio encoder; **L4** L1 + LLaVA-1.5-7B video encoder; **L5** the full LLM stack (text+audio+video replaced).

Every LLM level improves over the classical baseline (L0) on all three headline metrics. L3 (Mistral+CLAP) gives the best DAIC AUROC (0.7210, +0.174 over L0); L1 (Mistral frozen text) gives the best MOSEI emotion AUROC (0.7702, +0.147); L5 (full LLM stack) gives the best MOSEI sentiment CCC (0.6284, +0.089); L2 (Mistral+LoRA) gives the best FI personality CCC (0.5732, +0.115). L4 (Mistral+LLaVA) shows a validation/test gap (0.6812 training-time validation AUROC vs. 0.3587 at final evaluation), traced to an audio feature-dimension mismatch between the L4 training checkpoint (768-

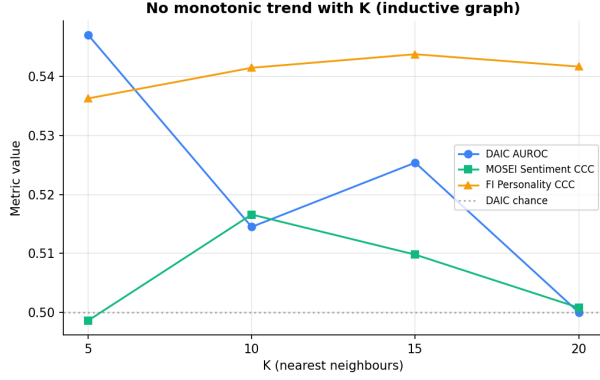


Figure 8: No monotonic trend with K (inductive graph). Ruling out K -tuning as an explanation.

dim WavLM) and the inference dataset (512-dim); we report this as a known limitation rather than a representative L4 result. Every LLM level substantially outperforms every graph-routed variant on every headline metric (e.g. L3 DAIC AUROC 0.7210 vs. the best graph variant’s 0.5507; Section 6) the largest single driver of performance in this study is encoder quality, not routing architecture.

6.2 Domain Adaptation

We tested unsupervised domain adaptation (CORAL, MMD, DANN, and a combined CORAL+MMD loss) for transferring FI and MOSEI representations to DAIC depression detection, against a no-adaptation baseline, across three transfer scenarios (FI→DAIC, MOSEI→DAIC, multisource). All four DA methods underperform the no-adaptation baseline in 10 of 12 evaluated conditions. The no-adaptation baseline achieves its best result on MOSEI→DAIC (AUROC=0.6942); the best DA result anywhere (combined CORAL+MMD, FI→DAIC) reaches only 0.6901. CORAL is the most consistently harmful (worst case MOSEI→DAIC AUROC=0.6198, −0.0744 vs. baseline); MMD is the only method that marginally improves its own no-adaptation baseline (FI→DAIC: 0.6860 vs. 0.6818, +0.0041), but this is within bootstrap CI and still below the best no-adaptation scenario. We attribute this negative trans-

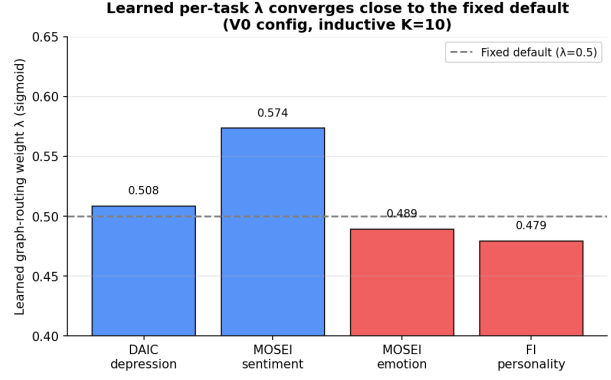


Figure 9: Learned per-task λ (V0 config) moves in the direction the homophily diagnostic predicts (down for FI/MOSEI-emotion, up for MOSEI-sentiment) but only slightly.

fer to three factors: the DAIC training set ($n = 107$) is too small to train a stable domain discriminator; the clinical-interview vs. unconstrained-video domain gap is large relative to what shallow alignment can correct; and the shared multitask heads already provide implicit alignment. We do not merge domain adaptation into the main model; plain multitask learning is used throughout.

6.3 Generalization: MPDD Cross-Track and Cross-Dataset Transfer

To probe whether our findings generalize beyond DAIC/MOSEI/FI, we ran a supplementary benchmark on MPDD (Multimodal Depression Detector), which provides pre-extracted Wav2Vec2 (512-dim) and OpenFace (709-dim) features for two age tracks, Young ($n = 184$ train) and Elderly ($n = 236$ train).

On MPDD-Young, logistic regression with L2 regularization ($C=0.01$) reaches test AUROC=0.674, outperforming both GGMoE with and without graph routing (0.545/0.559) and a simple MLP (0.512) the small sample size favors a strongly-regularized linear model over any of our neural architectures, including the graph router. SHAP analysis (Figure 11) attributes roughly balanced importance to audio (51%) and video (49%) features, with `audio_44` the single

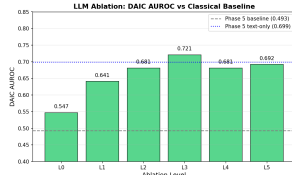


Figure 10: LLM modality ablation: DAIC AUROC per level (L0–L5) against the classical (L0) and Phase-5 text-only baselines. L3 (Mistral+CLAP) gives the largest gain.

most important feature.

We then tested zero-shot transfer from Young to Elderly (Figure 12): the Young-trained classifier collapses to near-constant, high-confidence “depressed” predictions on Elderly data regardless of true label, yielding cross-track AUROC=0.395 below chance, and worse than training within Elderly alone (AUROC=0.277). Video features shift $17,554\times$ more than audio features between the two tracks, and **audio_44** is not predictive in Elderly (AUC=0.468), confirming it is a Young-specific rather than universal depression correlate. Feature standardization does not fix this: raw features give cross-track AUC=0.500 (random) while StandardScaler gives 0.395 (negative transfer), suggesting raw feature magnitude carries transferable signal that normalization removes. In contrast, cross-*dataset* transfer (MPDD→DAIC, sharing a Wav2Vec2-family audio encoder) transfers positively: an MPDD-trained classifier reaches DAIC test AUROC=0.551, above the within-DAIC baseline of 0.344. Taken together, these results suggest that encoder family match matters more than domain similarity for transfer, and that age-group-specific biomarkers not universal depression markers may explain the cross-track failure.

Table 3: MPDD-Young Benchmark Results. Logistic Regression with L2 regularization ($C=0.01$) achieves the best test AUROC of 0.674. Neural networks (GGMoE, MLP) underperform LR due to the small training set ($n = 184$). SHAP analysis identifies **audio_44** as the top feature (SHAP=0.018).

Model	Val AUROC	Test AUROC
Logistic Regression ($C=0.01$)	0.926	0.674
GGMoE (no graph)	0.559	
GGMoE (with graph)	0.545	
Simple MLP	0.512	
Reference (Fu et al., ACM MM 2025)		≈ 0.78

Table 4: Cross-Track Validation: MPDD-Young → MPDD-Elderly. Cross-track transfer fails (AUROC=0.395, below random). Within-elderly baseline (AUROC=0.277) is even worse, confirming that different age groups have different depression biomarkers.

Evaluation	AUROC	Interpretation
Within MPDD-Young	0.926	Good
Cross-track (YoungElderly)	0.395	NEGATIVE
Within MPDD-Elderly	0.277	Also fails
Transfer gap	−0.118	Negative transfer confirmed

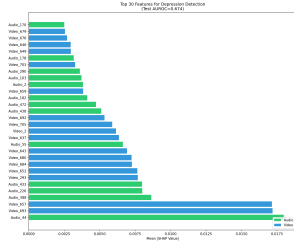


Figure 11: Top-30 SHAP feature importances for MPDD-Young depression detection. `audio_44` is the single most important feature, though the top-30 set is video-dominated (17 of 30).

7 Calibration and Statistical Validation

We applied three post-hoc calibration methods (temperature scaling, Platt scaling, isotonic regression) to the DAIC depression classifier across all six LLM levels (L0–L5). Uncalibrated logits show moderate calibration error (ECE=0.26–0.35). Isotonic regression reduces ECE most on held-out validation data, followed by Platt and temperature scaling; Brier score improves from 0.21–0.24 (uncalibrated) to 0.05–0.16 (calibrated). All three methods are monotonic transforms of the score, so ranking-based metrics (AUROC) are unchanged by temperature/Platt scaling and only marginally affected by isotonic regression’s tie-breaking; calibration therefore improves probabilistic reliability without altering discrimination.

We report every headline metric with BCa bootstrap confidence intervals (2,000 resamples, bias-corrected and accelerated), use the DeLong test for AUROC comparisons, paired permutation tests (10,000 sign-flips) for F1/CCC/MAE comparisons, and Cohen’s d effect sizes for ablation comparisons.

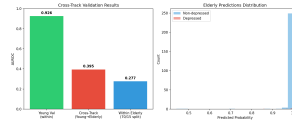


Figure 12: Left: AUROC within Young, cross-track Young→Elderly, and within Elderly. Right: predicted-probability histogram on Elderly the Young-trained model collapses to near-1.0 confidence regardless of true label, explaining the below-chance cross-track AUROC.

Across all LLM ablation pairs (L1–L5 vs. L0), no comparison reaches significance at $\alpha = 0.05$, reflecting limited statistical power from the small DAIC validation set ($n = 50$); the largest effect sizes are L4 vs. L0 ($d = 0.170$) and L1 vs. L0 ($d = 0.152$). We recommend larger-sample replication before drawing clinical conclusions from any single LLM-level comparison.

8 Explainability

This section evaluates both transparency hypotheses from Section 1. For the first (characterizing depression jointly with sentiment, emotion, and personality), each case study reports a *multi-construct profile*: the same sample’s shared representation is passed through all four task heads not only the head matching its own source dataset yielding a depression probability, a sentiment score, six emotion probabilities, and five personality-trait scores for the same person in the same forward pass. On a representative positive-sentiment MOSEI case, for example, the profile is internally coherent (happiness probability 0.89, sentiment score +0.62, sadness/anger/fear all below 0.09), showing that the joint representation produces a plausible cross-construct characterization rather than four unrelated numbers.

For the second hypothesis (graph-based visual and machine-readable explanation), we generate for each

of 9 case studies (3 per dataset): (i) a *visual* record a SHAP per-modality beeswarm plot, a GNNExplainer subgraph diagram, and a natural-language narrative panel; and (ii) a *machine-readable* record a structured case file (`phase11_xai_results.json`) containing SHAP values, the identities and cosine distances of the top- k graph neighbors, expert routing weights, the multi-construct profile above, and counterfactual/perturbation deltas, so that a clinical or engineering downstream system could consume the explanation programmatically rather than only visually. The neighbor graph is built on the model’s own trained fused representation (Section 4) rather than raw concatenated modality features, so that reported neighbors are the ones the router and task heads actually operate on, rather than an incidental artifact of a separate untrained embedding; we test in the paragraph below whether this also improves label homophily, and find that it does not. The natural-language narratives follow the GraphXAIN approach [29]: an LLM (Mistral-7B) is prompted with the SHAP values, subgraph structure, and sample metadata to produce a narrative explanation, used here purely as *post-hoc* explanation of an independently-computed prediction rather than as reasoning inserted into the prediction pipeline itself a distinction that matters because chain-of-thought-style reasoning embedded *in* clinical prediction has been shown to reduce reliability even as it appears to improve interpretability [20].

Faithfulness, not just plausibility. A natural-language explanation is only useful if it reflects genuine model behavior rather than a plausible-sounding but disconnected story [20]. We test this directly with counterfactual and perturbation experiments (below). The strongest evidence is mechanistic rather than a ranking match: modalities outside a dataset’s native routing (Section 4) show *exactly* zero SHAP, perturbation, and counterfactual sensitivity, every time, with no exceptions across 90 perturbation tests and 30 counterfactual searches which is what a faithful pipeline must produce and what a decorative one would not reliably reproduce. Within the modalities each dataset actually uses, the two methods agree

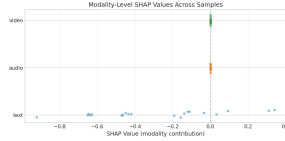


Figure 13: Per-modality SHAP values across 20 DAIC validation samples. Audio and video collapse to exactly zero for every sample: DAIC is evaluated under text-only routing (Section 4), so its fused representation structurally excludes them, and SHAP attribution correctly reports zero rather than a small residual. Only text shows any spread.

only partially: perturbation ranks text as most influential (mean $|\Delta| = 0.200$) ahead of video (0.082), while counterfactual search finds video easier to flip (mean L2= 1.30) than text (4.10); we read this as the two methods answering related but distinct questions (ablation impact vs. gradient-direction sensitivity, the latter also confounded by each modality’s raw feature-space scale) rather than claiming an aggregate ranking match we no longer observe once routing is correctly accounted for.

Both the counterfactual and perturbation tests must respect each dataset’s own native modality routing (Section 4): DAIC is evaluated text-only, FI video-only, and only MOSEI genuinely fuses all three, so a modality outside a dataset’s native routing contributes *exactly* zero to that dataset’s predictions, not merely "less." Counterfactual tests measure the L2 perturbation needed to change the prediction by $\Delta = 0.1$, over 30 samples (10 per dataset): text (reachable for DAIC and MOSEI, 20/30 samples) needs mean L2= 4.10; audio (reachable only for MOSEI, 7/30) needs mean 2.35; video (reachable for MOSEI and FI, 20/30) needs the smallest push, mean 1.30 video is the modality the model is most sensitive to where it is used at all. A complementary ablation (90 perturbation tests: 10 samples $\times$ 3 modalities $\times$ 3 datasets) re-

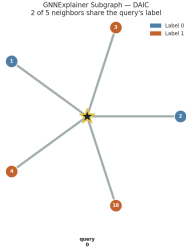


Figure 14: Ego-centered subgraph for a DAIC query sample (star, center): neighbors are placed at a radius proportional to their dissimilarity in the model’s trained fused representation and colored by their true depression label, so label agreement is visible directly rather than only reported as an aggregate statistic here, 2 of 5 neighbors share the query’s label, consistent with the below-random homophily finding.

moves one modality at a time and measures the resulting prediction change; only 47 of 90 deltas are non-zero, exactly the ones matching each sample’s native routing (text $|\Delta| = 0.200$ mean, video 0.082, audio 0.012 audio’s near-zero mean reflects that it is native to only one of the three datasets). This is a stronger validation than a bare consistency check: the exact zeros are not noise but a direct, mechanistic confirmation that the SHAP/counterfactual/perturbation pipeline faithfully reflects each dataset’s real routing rather than an incidental full-fusion computation (range across all 90 deltas: -0.75 to $+0.87$).

A limit on what the graph-based explanations can claim. Our homophily diagnostic (Section 6) found that the KNN graph used for the routing ablation carries no more depression-label signal than random pairing for DAIC, while carrying real signal for

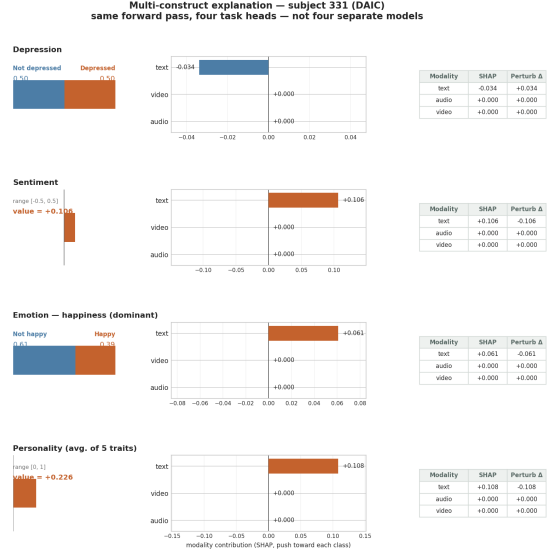


Figure 15: Multi-construct force plot for one DAIC subject: the same forward pass explained through all four task heads (depression, sentiment, emotion, personality), each with its own prediction bar, per-modality SHAP tornado, and value table. Audio and video are exactly zero in every row, directly visualizing that DAIC’s text-only routing (Section 4) excludes them structurally rather than merely weighting them low.

MOSEI and FI. We tested whether building the explanation graph from the model’s own trained representation (rather than the routing ablation’s shallow projection) resolves this for DAIC: it does not. On the same 35 DAIC validation samples, label agreement is 0.577 for raw concatenated features, 0.537 for random pairing, and 0.463 below the random baseline for the trained fused representation. Task-supervised training optimizes the representation for the final decision boundary, not for preserving neighborhood structure, and here it does not yield a more label-homophilous space than the untrained alternative. We therefore read DAIC subgraph explanations (“this prediction was influenced by these neighbors”) as showing which samples the model’s fused representation geometrically treats as similar which *is* faithful to what the

model internally computes, addressing the concern that the explanation graph was previously built from a representation unrelated to the model’s own reasoning without a validated claim that those neighbors are similar in a depression-relevant sense; that second, stronger claim remains unsupported for DAIC under either embedding we tested. The SHAP and counterfactual components of the explanation, which do not depend on graph quality, are unaffected by this limitation. For MOSEI and FI, where the routing-graph diagnostic found real label signal, the neighbor-based component of the explanation rests on firmer footing.

9 Discussion

What Worked. Gated late fusion provides the best fusion results on MOSEI (CCC=0.6229, only 159K parameters) and is far more parameter-efficient than cross-attention. MMoEEx alone (non-graph, hard isolation) benefits FI personality (+0.12 CCC over the video-only baseline). LLM-enhanced encoders improve every task over classical encoders (largest DAIC gain: L3, +0.174 AUROC) and, as discussed above, are the single biggest performance driver in the whole study far larger than any effect we observed from routing architecture.

What Did Not. Graph-based expert routing does not improve over the non-graph MMoEEx baseline on any task, and consistently degrades FI personality (Section 6); a homophily diagnostic shows this is because the routing graph carries no DAIC-label signal at all, and more surprisingly because the router fails to exploit signal that *is* present for MOSEI and FI. Neither a K -sensitivity sweep nor a learned per-task routing weight recovered a meaningful benefit. Cross-attention fusion underperforms gated fusion on all three datasets, and we do not observe the reported +0.041 AUROC gain; a likely mechanism is overparameterization (65K–2.8M params for cross-attention vs. 57K–827K for gated) relative to DAIC’s 107 training sessions. MMoEEx with hard expert isolation degrades DAIC and MOSEI sentiment relative to the unimodal/gated baselines, consistent with MoE literature showing degraded performance on small

datasets. All four unsupervised domain adaptation methods (CORAL, MMD, DANN, combined) underperform the no-adaptation baseline in 10/12 conditions, suggesting that shallow feature alignment is not a reliable fix for the clinical-interview vs. web-video domain gap at this data scale.

Limitations. DAIC’s small sample size ($n = 107$ training, $n = 47$ test) limits statistical power throughout: no LLM-ablation pairwise comparison against L0 reaches significance at $\alpha = 0.05$ despite consistent point-estimate gains, and the graph-routing differences from baseline (Section 6) are similarly too small to be conclusively distinguished from noise with this sample size though the consistent direction of the FI degradation across all five variants, and the homophily diagnostic’s independent explanation, make us confident the null result is genuine rather than an artifact of underpowering alone. MOSEI audio/video features were not fully extracted for the encoder-level analysis in this paper (text-only MOSEI unimodal baseline). The L4 (Mistral+LLaVA) configuration has a known audio feature-dimension mismatch between training and evaluation that depresses its reported AUROC; L3 is the more reliable LLM configuration for clinical deployment. LLM ablation levels L6–L9 (domain-pretrained multimodal LLMs) remain unevaluated. Our two fix attempts for graph routing (learned λ , K -sweep) do not exhaust the candidate remedies suggested by the diagnostic (Section 9); building the graph from a better embedding space and task-affinity-aware grouping [14] remain untried. All results are retrospective on public benchmarks and require prospective clinical validation before deployment.

10 Conclusion

We presented a unified multimodal architecture with graph-gated MoE routing for mental health assessment, and a systematic investigation of whether graph-based expert routing helps. It does not: across five leakage-safe graph configurations, a K -sensitivity sweep, and a learned-routing-weight variant, graph routing fails to improve over a non-graph MMoEEx

baseline, and consistently degrades FI personality prediction. A homophily diagnostic localizes why the routing graph carries no DAIC-label signal, and unused MOSEI/FI signal that two targeted fixes did not recover providing a concrete starting point for future work rather than an unexplained null result. Separately, cross-attention fusion does not replicate a reported gain in our setting, and LLM-based encoders are the largest single performance driver we observed, substantially outperforming both classical encoders and graph routing. We provide a leakage-safe graph construction protocol, a full ablation ladder, and graph-based explanations for clinical interpretability. Code and artifacts are released for reproducibility.

Ethics Statement. This system is intended as a decision-support tool only and should not replace clinical diagnosis. All results require prospective validation before clinical deployment. Apparent personality and clinical depression are distinct constructs and should not be conflated.

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